

The Recursion Call Tree

Calls go DOWN until the base case. Then answers come back UP. Nothing returns until the bottom is reached.

```
function factorial(n) {  
  if (n <= 1) return 1; // base case  
  return n * factorial(n - 1); // recursive case  
}
```

Every recursive function needs two things

1. A BASE CASE that returns without calling itself again.
 2. A step that moves CLOSER to it: here, $n - 1$.
- Miss either one and every call adds another frame until the engine gives up:
RangeError: Maximum call stack size exceeded.

Recursion shines where the DATA is nested: a folder tree, a DOM subtree, a JSON document. For a flat list, a loop is simpler and cannot blow the stack. Each pending call holds a real stack frame, which is why depth, not total work, is the limit.

